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Do Geopolitical Risk Factors and Economic Policy Uncertainties Stimulate Capital Outflows? An Updated Evidence.

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ABSTRACT

Most developing countries are making frantic efforts to curtail illicit outflows of investible funds. Unfortunately, these efforts do not seem to have produced the expected results. Meanwhile, existing empirical investigations have failed to explain emphatically the channels through which this economic sabotage is being perpetuated. In this context, this study scrutinizes whether geopolitical risk factors (GPR) and economic policy uncertainty (EPU) stimulate the outflows of illicit capital from South Africa over 1985 – 2018 while controlling for the influence of national income, trade openness, foreign direct investments (FDI) and population. Preliminary investigations confirmed cointegration among the series. Furthermore, capital flight is a systemic problem considering its significant short-run autoregressive effect. Further evidence from the ARDL model indicates that GPR and EPU exert significant positive impacts on capital flight only within the short run, whereas FDI produces a positive and significant influence on capital outflows at all times. The influence of FDI on illicit capital outflow is also consistent when the Kernel Regularized Least Squares (KRLS) model was applied, such that its positive and significant influence was also observed both on the average and across all quantiles of the distributions of capital migrations. Therefore, to curtail capital flight, policymakers must keep an eye on the inflows and outflows of FDI. Meanwhile, GPR and EPU are not long-term predictors of capital migrations from South Africa. The findings of this investigation could be equally beneficial to other countries battling capital flight.

Keywords: Capital flight; Geopolitical Risk Factors; Economic Policy Uncertainty; Kernel Regularized Least Squares; South Africa.

JEL Classification: C23, F21, F30, F32, F45, O40, 055

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¿Los Factores de Riesgo Geopolítico y la Incertidumbre de la Política Económica Estimulan las Salidas de Capital? Una Evidencia Actualizada.

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RESUMEN

La mayoría de los países en desarrollo están haciendo esfuerzos frenéticos para reducir las salidas ilícitas de fondos invertibles. Por desgracia, estos esfuerzos no parecen haber dado los resultados esperados. Entretanto, las investigaciones empíricas existentes no han logrado explicar con rotundidad los canales a través de los cuales se está perpetuando este sabotaje económico. En este contexto, este estudio examina si los factores de riesgo geopolítico (GPR) y la incertidumbre de la política económica (EPU) estimulan las salidas de capital ilícito de Sudáfrica entre 1985 y 2018, controlando al mismo tiempo la influencia de la renta nacional, la apertura comercial, las inversiones extranjeras directas (IED) y la población. Las investigaciones preliminares confirmaron la cointegración entre las series. Además, la fuga de capitales es un problema sistémico teniendo en cuenta su significativo efecto autorregresivo a corto plazo. Otras pruebas del modelo ARDL indican que el GPR y el EPU ejercen efectos positivos significativos sobre la fuga de capitales sólo a corto plazo, mientras que la IED produce una influencia positiva y significativa sobre las salidas de capitales en todo momento. La influencia de la IED en la salida de capitales ilícitos también es consistente cuando se aplica el modelo de mínimos cuadrados regularizados de Kernel (KRLS), de forma que su influencia positiva y significativa también se observó tanto en la media como en todos los cuantiles de las distribuciones de las migraciones de capital. Por lo tanto, para reducir la fuga de capitales, los responsables políticos deben vigilar las entradas y salidas de IED. Mientras tanto, el GPR y el EPU no son predictores a largo plazo de las migraciones de capital desde Sudáfrica. Las conclusiones de esta investigación podrían ser igualmente beneficiosas para otros países que luchan contra la fuga de capitales.

Palabras clave: Fuga de capitales; Factores de Riesgo Geopolítico; Incertidumbre de Política Económica; Mínimos Cuadrados Regularizados de Kernel; Sudáfrica.

Clasificación JEL: C23, F21, F30, F32, F45, O40, 055

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1. Introduction

Notably, international communities and other relevant agents provide various dimensions of assistance to Africa and other developing countries. Such packages include both financial and nonfinancial assistance. However, it is doubtful if these international donor agencies consider the weight of damage the continuous outflow of illicit funds from the continent has on the development of these developing countries. On their part, the African countries, South Africa inclusive, bearing the heaviest burden of capital migrations must not relent on the fight against this notable economic sabotage. However, without a concerted commitment to understanding the relevant enablers, their efforts remain futile. Understandably, several factors, such as income (Ndikumana, 2016), exchange rate (Pradhan & Hiremath, 2020; Uche & Effiom, 2021a), globalization (Rahman, Mustafa & Turpin, 2019), greed (Ndoricimpa, 2018), foreign aid (Aziz, 2014; Collier, Hoeffler & Pattillo, 2001), terrorism (Ndiaye & Siri, 2016), external debt (Ramiandrisoa & Rakotomanana, 2016), exchange rate devaluation (Asongu & Nnanna, 2020), foreign direct investment (Ndikumana & Sarr, 2016; Ndikumana & Sarr, 2019), corruption (Osei-Assibey, Domfeh & Danquah, 2018), economic policy uncertainties (Salisu & Isah, 2021; Uche & Effiom, 2021a), and institutions (Asongu & Nnanna, 2020); Rahman, Mustafa & Turpin, 2019), among other factors, have been identified as major culprits leading to the continued outflows of illicit funds from developing to developed nations. However, the lack of consensus and the ambiguities in these existing studies cast doubt on their ability to provide streamlined policy strategies to mitigate the economic sabotage faced by these developing countries. Hence, further investigations in this regard are passionately appealing.

Conceptually, capital flight and illicit financial outflows are flawed by a lack of terminological clarity and hence are used interchangeably. Whereas Reuter (2012) concedes that capital flight is more about the outflows of money or capital from a nation with the purpose of finding a more secure investment environment and creating a higher profit, he defines illicit financial flows as flows that cause damage to the economic development of the country. UNESCO (2013) asserts that capital flight may be legal or illegal. This position suggests that all financial outflows presage capital flight. (Global Financial Integrity, [GFI] 2017) refers to illicit financial outflow as the illegal movement of money or capital from one country to another, while the African Development Bank, [AFDB]2016) conceptualizes illicit financial flows as money illegally earned, transferred or used. From the foregoing, our study adopts the use of illicit financial flows (IFFs) to connote the illegal movement of funds abroad, including aid, investment, trade, debt, health tourism, migrant remittances and foreign reserves, to secure better profits due to an unfavourable business environment in the country of origin. We adjudge these funds movement as illegal since they are not captured and recorded by the balance of payment statistics.

Regarding the theoretical link between capital flight and economic fundamentals, Ajayi (1997) and Pastor (1990) consider debt-driven capital flight or debt overhangs, investment diversion, and austerity and tax-depressing dynamics. The debt revolving door hypothesis is also suggested by Gankou et al. (2016). Unarguably, capital flight from developing countries mirrors the Lucas paradox (Lucas, 1990), which was exposed further in Ndikumana (2016) and Uche & Effiom (2021a). The Lucas paradox synthesizes an ironic and pathetic situation where countries with a notable dearth of financial resources but with potentially high returns instead of investing such funds reap the possible high returns rather than starving the economy of such financial resources by migrating these funds to more prosperous nations where such funds are rarely appreciated. The liquidity constraint or crowding out effect may result in depreciation of the domestic currency if the authorities are operating a floating exchange rate system (Ajayi, 1997). An attempt to defend the exchange rate at this time leads to the loss of international reserves. The investment diversion thesis provides one of the well-known negative consequences of capital flight in the countries involved.

Empirical expositions on the determinants of capital flight cut across several economies, including studies focusing on Africa and some other developing countries. Some studies are country specific, while some are panel data studies. Accordingly, Uche & Effiom (2021a) expose a disconnection between EPU and capital flight in Nigeria within a QARDL setting. A divergent submission based on the MG and PMG estimator was advanced by Salisu & Isha (2021) in terms of a significant influence of macroeconomic uncertainties on capital migrations in sub-Saharan African countries. Furthermore, Ndikumana & Sarr (2019) clarify that FDI fuels capital flight in a study of 30 African countries over 1970-2015. However, for Ndikumana & Sarr (2016), FDI has an insignificant influence on capital flight in 37 African countries between 1970 and 2013. Likewise, Mwangi, Njuguna, and Achoki (2019) submit that a surge in the current level of FDI leads to higher capital flight in Kenya, whereas a surge in past volume reduces FDI based on evidence from the ARDL procedure. Furthermore, Gankou et al. (2016) infer that FDI does not drive capital flight in Cameroon. For Anetor (2019), economic growth and exchange rates rather than trade openness and inflation are the main drivers of illicit capital outflows in sub-Saharan Africa. Pradhan and Hiremath (2020) further emphasized the significant influences of GDP and exchange rate on capital migrations from India. However, by applying the OLS technique in the study of the determinants of capital flight in Indonesia between 2009 and 2017, Basorudin & Kusmaryo (2018) highlight that GDP, exchange rate and inflation are not significant drivers of capital outflows. Forson, Obeng & Brafu-Insaidoo (2017), based on evidence from ARDL estimation, also contend that economic growth and financial development mitigate the rate of capital migrations significantly in Ghana.

Beyond the conflicting assertions and the notable ambiguities in existing studies on account of the relevant predictors of capital migrations, another major fact compelling this study is the lack of consideration of the potential influence of geopolitical risk factors (GPR) and economic policy uncertainties (EPU) on capital migrations, as witnessed in existing studies. Yes, we envisage that these factors (GPR) and (EPU) arising primarily from political and economic shocks could form formidable conduits for the persistent illicit outflows of investible funds from the continent. However, this assertion remains questionable and subjective until it is empirically verified. On this note, the current study is guided by the following objectives: to critically ascertain whether GPR and EPU exacerbate illicit capital outflows; to understand whether the trend of capital migrations is consistent or heterogeneous over the distributions; and to provide updated evidence on the critical roles of income, trade openness, foreign direct investment (FDI) and population in capital flight. Consequently, the following empirical questions are relevant in this study: what role do GPR and EPU play in capital migration? Are the effects of GPR and EPU on capital migrations consistent at all times? What effects do the changing trends of income, trade openness, FDI and population have on illicit capital migrations? Expectedly, the provision of evidence-based answers to these questions would, to a very large extent, avail policy strategies to ensure that capital migrations are mitigated, particularly in South Africa, as well as other developing countries contending with this economic woe.

Among other African countries battling the menace of illicit financial outflows, South Africa, a notable economic giant in Africa, is among the top five countries (the second coming behind Nigeria) with a notorious record of capital migrations. Specifically, South Africa lost more than $100 billion from 1970 to 2018 (Ndikumana & Boyce, 2021). According to records, capital flight from South Africa in 2018 represents approximately 89.5% of total GDP. Notably, the volume of capital flight from these countries, including South Africa, exceeds the volume of other flows, including remittances, official debt assistance (ODA), foreign direct investment (FD) and portfolio investment (Ndikumana & Boyce, 2021). Notably, South Africa remains the choice destination for policy experimentations with the option of disseminating such strategies to other African countries. Therefore, the current study believes that empirically exposing the impasse of GPR and EPU on capital migrations from South Africa is of immense benefit for curtailing the illicit outflows of funds from other African countries and other developing countries on other continents.

At this point, it is pertinent to highlight the important contributions of this study by specifically highlighting how it extends the trajectories of existing knowledge. First, to the best of our knowledge, no existing study has considered the influence of GPR and EPU on illicit capital migration in the context of any African country or South Africa in particular. Because of the enlisted control variables, existing studies relied on non-updated datasets for their empirical analysis, which resulted in their unconvincing assertions. Unlike previous studies, this investigation relies on the most updated capital flight series available at the Political Economic Research Institute (PERI) for empirical expositions. In addition, the application of the newly introduced kernel regularized least squares (KRLS) econometric algorithm marks a clear extension of the trajectories of existing knowledge in terms of studies investigating the relevant determinants of illicit capital outflows both in South Africa and elsewhere. Notably, the KRLS process is a flexible machine learning technique that accounts for the interactions of the regressor(s) and the regressand both on the average and across the quantile distributions of the explained variable. As such, the possible heterogeneous interferences are accounted for by considering the inability of the dominant techniques in existing studies to empirically elicit such vital information. As expected, this holistic investigation embellished with enhanced econometric algorithms will provide streamlined policy strategies to curtail the outflows of investible funds from South Africa. Likewise, such policy strategies could also avail other African countries, as well as other developing countries relevant pathways to mitigate capital flight.

Going forward, the relevant information on the analysed datasets and empirical models/methodologies are exposed in section 2, followed by empirical estimations and discussions of the outcomes in section 3. Meanwhile, the study concludes with some relevant policy briefs in section 4.

2. Data and methodology

Data

The study relies essentially on annual frequency datasets between 1985 and 2018 for empirical investigations. Meanwhile, the datasets were extracted from various public data repositories notable for their reliability. This includes the Political Economy Research Institute (PERI) of the University of Massachusetts, Amherst[1], for capital flight; the world uncertainty index compiled by Ahir, Bloom & Furceri (2018)[2], a proxy for EPU; and the geopolitical risk index computed by Caldara & Lacoviello (2021) for the GPR dataset[3]. Furthermore, the other enlisted control variables, including national income (GDP growth – annual %), trade openness (trade, % of GDP), FDI (net inflow, % of GDP), and population (total), were sourced from world development indicators (WDI) of the World Bank. Meanwhile, to ensure consistency in data frequency, the datasets on GPR and EPU were converted to annual series through date with frequency conversion (DFC) following previous studies, including Godil et al. (2021) and Uche & Effiom (2021b). For consistency, datasets on capital flight, GPR, trade openness and population were converted to natural logarithmic values such that the estimated results are reported as elasticities. Graphical illustrations of the relevant series are provided below for more insights into the dynamics of GPR, EPU and capital flight in South Africa. The trends suggest that capital flight has been increasing since 2005 with a significant drop in 2015, which further rose in 2017 with a noticeable decline in 2018. There is evidence of opposing effects in terms of GPR and EPU such that GPR has been declining since 2005, whereas EPU has been increasing since 2011.

[1] https://www.peri.umass.edu/capital-flight-from-africa
[2] https://www.policyuncertainty.com/infectious_EMV.html
[3] https://www2.bc.edu/matteo-iacoviello/gpr.htm.

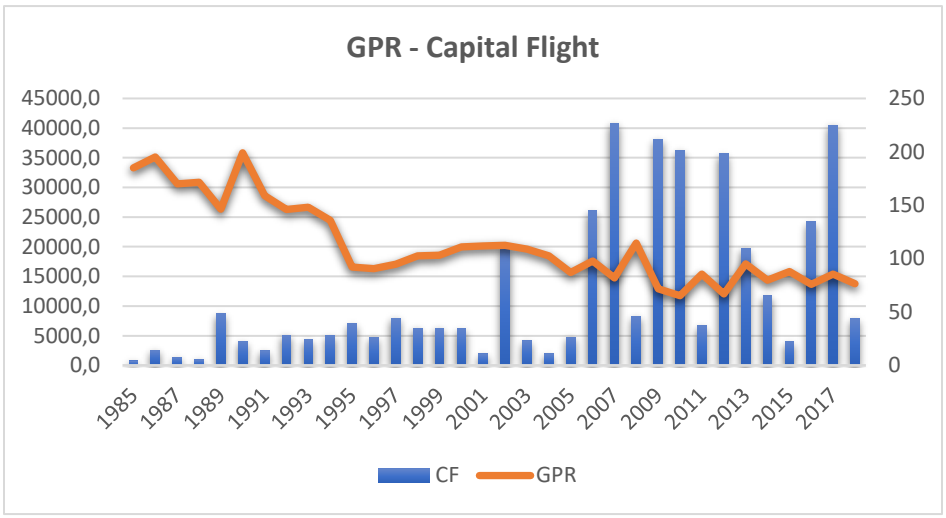


Fig. 1: Capital flight – geopolitical risk dynamics

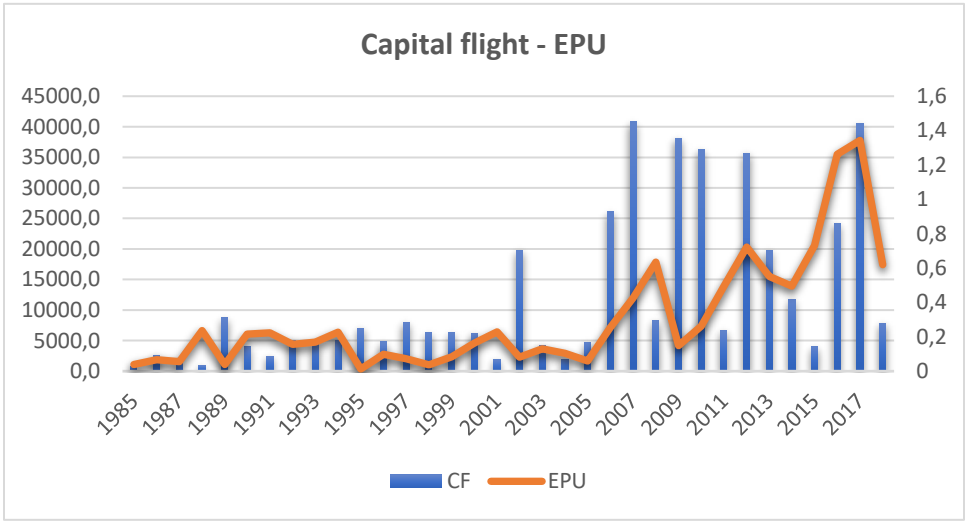


Fig. 2: Capital flight – economic uncertainty dynamics

Model

To provide in-depth empirical illustrations following the objectives of the study, that is, to ascertain the potential influence of GPR and EPU on illicit outflows of investible funds from South Africa and the moderating imperatives of income, trade openness, FDI and population. It is sufficient to highlight that the inclusion of the enlisted control variables is justified by previous related studies (see Anetor, 2019; Asongu & Nnanna, 2020). Moreover, conforming to existing studies (Effiom, Uche & Otei, 2021; Salisu & Isah, 2021; Uche & Effiom, 2021a), the baseline equation for this investigation is provided:

$$lnCF = (lnGPR, EPU, GRT, lnOPN, FDI, lnPOP) \quad (1)$$

Accordingly, lnCF, lnGPR, EPU, GRT, lnOPN, FDI and lnPOP symbolize the natural logarithm value of capital flight, geopolitical risk factor, economic policy uncertainty, economic growth, trade openness, foreign direct investment and population, respectively. The extended econometric version of the model is provided below (Eq. 2) for more insights.

$$lnCF_t = \beta_0 + \beta_1 lnGPR_t + \beta_2 EPU_t + \beta_3 GRT_t + \beta_4 lnOPN_t + \beta_5 FDI_t + \beta_6 lnPOP_t + \varepsilon_t \quad (2)$$

In Eq. 2, all the datasets retained their previous definition with the inclusions of t that indicate time-series dimensions and the relevant coefficients (β0, β1, β2, β3, β4, β5, β6) for each parameter, while ε_t represents the stochastic error term. Considering that extant theories did not provide an explicit account of the possible effects of the enlisted variable on capita flight, we expect further clarifications from the empirical outcomes of the present study.

Estimation techniques

As previously stated, the study adopts two dynamic and flexible econometric algorithms, including the autoregressive distributed lag (ARDL) akin to Pesaran & Shin (1999) and the kernel regularized least squares (KRLS) attributable to Hainmueller and Hazlett (2014) models for its empirical outcomes. Specifically, the unique attributes of the ARDL include the fact that it accounts for both short-run and long-run impacts, and it allows for mixed integrated data of order zero and order one. It also identifies the possible speed of adjustment to equilibrium after initial short-run perturbations (Omoke & Uche, 2020; Omoke & Uche, 2021; Sarkodie & Owusu, 2020; Uche et al, 2021; Uche et al., 2022). In this part, the KRLS, a flexible machine learning nonlinear algorithm, is also unique considering that it provides information on the possible impacts at the average and across quantile distributions of the explained variable. It also accommodates both stationary and nonstationary datasets (Choi & Lee, 2020; Sarkodie & Owusu, 2020). As expected, the combinations of these two unique models will provide for comparisons and ensure in-depth and reliable inferences that will invoke policy strategies to mitigate the illicit outflows of investible funds from South Africa. Furthermore, the typical pathways of the ARDL and KRLS models are provided herewith:

$$ln\Delta\gamma_t = \beta_0 + ln\beta_1\gamma_{t-1} + ln\beta_2 x_{t-1} + \sum_{i=1}^{n} \theta_1 ln\Delta\gamma_{t-i} + \sum_{i=0}^{n} \theta_2 ln\Delta x_{t-i} + \varepsilon_t \quad (3)$$

where γ_t signals the dependent variable, x_t indicates the vector of the independent variable(s), Δ is the difference operator, *ln* represents the natural log function for each variable, and ε_t is the stochastic error component. $\sum_{i=1}^{n} \varphi_i ln\Delta\gamma_{t-i}$ signals the short-run framework, and $\beta_1\gamma_{t-1}$ indicates the long-run equilibrium specification. Going by the above specification, the study will reveal both the contemporaneous and long-term impasse of GPR, EPU and the enlisted control variables on illicit outflows of investible funds from South Africa, thereby unveiling relevant policy pathways to forestall future occurrences. Aside from the application of the linear model (ARDL) that provides for both long- and short-run interactions, the study extends the trajectory of knowledge by adopting a heterogeneous nonlinear model (KRLS) process to account for the impact of the identified explanatory factors on capital outflows in South Africa both on average and at different quantiles of capital flight distributions. Consequently, it is expected that the empirical revelations of this extended model provide broad-based policy imperatives that would, to a very large extent, ensure the mitigation of capital migrations from the country. Against this background, we therefore provide the typical pathways of the KRLS estimator following Choi & Lee (2020) and Wang, Cheng & Qi (2020).

$$f(x_j, x_i) = exp\left(-\frac{|| x_j - x_i ||^2}{\sigma^2}\right) \quad (4)$$

where $|| x_j - x_i ||^2$ is the Euclidean distance between the independent variables x_i and x_j, and $\sigma^2 \in R^+$ denotes the bandwidth of the kernel specification.

$$y^* = f(x^*) = \sum_{i=1}^{N} = c_i k\ (x^*, x_i) \tag{5}$$

where the objective function $f(x^*)$ is regarded as a linear combination of many kernel specifications $k(x^*, x_i)$, and c_i is a weight for each covariate vector. Since $k(x^*,x_i)$ is the weight of the similarity between x^* and x_i, we observed that the value of $k\ (x^*, x_i)$ is expected to grow larger as we move the test point x^* much closer to x_i. Furthermore, the expected outcomes at the test point are provided by a weighted sum of how similar the result relates to each observation in the (training) dataset. Similar to the principle of the generalized linear model, the estimate can apply a set of kernel functions that signal the similarity of the sample observations to replicate the natural measure of the data. In that case, for any explained variable x, there should be a weight vector c_i(i = 1, …, N), which allows a linear fitting function of the mathematical expected value of the dependent variable to be specified based on the similarity of the explanatory variable x to the observed value, as provided in equation 6:

$$y = f(x) = \sum_{i=1}^{N} = c_i k\ (x, x_i) \tag{6}$$

3. Results and discussions

The empirical analysis of this study streams from a wide range of pre-estimation tests, which include summary statistics, correlation tests, stationarity tests, and cointegration tests. In the same vein, the outcomes are also validated based on several postestimation diagnostic tests, including tests for autocorrelation, heteroscedasticity, normality, specifications and stability. Furthermore, Table 1 projects the outcomes of the summary statistics and the collinearity test.

Table 1. Summary statistics and correlation matrix

Coefficients	$lnCF_t$	$lnGPR_t$	EPU_t	GRT_t	$lnOPN_t$	FDI_t	$lnPOP_t$
Panel A: Summary statistics							
Mean	8.85	4.68	0.31	2.18	3.95	1.02	7.65
Median	8.74	4.63	0.19	2.44	3.94	0.65	7.66
Maximum	10.61	5.29	1.34	5.60	4.28	5.98	7.76
Minimum	6.61	4.18	0.01	-2.14	3.62	-0.76	7.51
Std. Deviation	1.08	0.32	0.33	2.01	0.16	1.31	0.07
Skewness	0.03	0.43	1.75	-0.35	-0.28	1.83	-0.31
Kurtosis	2.36	2.16	5.66	2.45	2.39	7.29	2.04
Jarque-Bera	0.57(0.74)	2.08(0.35)	27.33***	1.13(0.56)	0.98(0.61)	45.13***	1.86(0.39)
Panel B: Correlation matrix							
Series	$lnCF_t$	$lnGPR_t$	EPU_t	GRT_t	$lnOPN_t$	FDI_t	$lnPOP_t$
$lnCF_t$	1.00						
$lnGPR_t$	-0.72***	1.00					
EPU_t	0.48***	-0.45***	1.00				
GRT_t	0.14	-0.29*	-0.09	1.00			
$lnOPN_t$	0.47***	-0.57***	0.54***	0.32*	1.00		
FDI_t	0.24	-0.43**	0.13	0.23	0.45***	1.00	
$lnPOP_t$	0.69***	-0.88***	0.67***	0.23	0.70***	0.45***	1.00

Note: For Panel A, *** indicates that the variable deviates from normalcy at the 1% significance level. For Panel B, ***, ** and * suggest correlations between the variables at the 1%, 5% and 10% significance levels, respectively.

The outcome of the summary statistics (Panel A – Table 1) indicates a convergence between the mean, median, maximum and minimum values of all the enlisted variables, which technically points to a reliable estimation. The standard deviation suggests that national income (GRT_t) and FDI_t are more dispersed than other variables. Accordingly, *lnCF, lnGPR, EPU* and *FDI* have positive tails, whereas *GRT, lnOPN* and *lnPOP* have negative tails. Judging by the kurtosis and Jarque-Bera statistics, all the series are normally distributed except *EPU* and *FDI*. The overall outcome of the summary statistics is indicative of a potential robust estimation. Furthermore, the correlation matrix (Panel B – Table 1) indicates a significant negative correlation between *GPR* and capital flight, whereas the matrix indicates a significant positive correlation between *EPU*, openness, population and capital migration. To provide more evidence in terms of the characteristics of the enlisted variables and possible long-run interactions, we outlined the unit-root and cointegration test results in Tables 2 and 3, respectively.

Table 2 Unit-root test

	Augmented Dickey-Fuller (ADF) test		Lee-Strazicich (LS) test			
Series	ADF Level	ADF First Diff.	LS Level	Break Dates	LS First Diff.	Break Dates
$lnCF_t$	-1.98	-8.53***	-5.99***	1993	-	-
$lnGPR_t$	-1.75	-9.98***	-3.18	1990	-4.26***	2014
EPU_t	1.62	-6.48***	-4.55***	2007	-	-
GRT_t	-3.23**	-	-4.93***	2000	-	-
$lnOPN_t$	-1.28	-5.43***	-2.02	1989	-5.88***	1990
FDI_t	-4.53***	-	-5.48***	1994	-	-
$lnPOP_t$	-1.56	-3.09**	-3.99**	2002	-	-

Note: *** and ** reveal that the series is stationary at the 1% and 5% significance levels, respectively.

Table 3 – Cointegration (Bounds) Test

Model	F-Statistics	Lower Bounds *I(0)*	Upper Bound *I(1)*	Remarks
ARDL				
lnCF = f(lnGPR, EPU, GRT, lnOPN, FDI, lnPOP)	**7.53*** **			Cointegrated
	Critical Values			
	1%	2.88	3.99	
	5%	2.27	3.28	
	10%	1.99	2.94	

Note: *** indicates cointegration among the series at the 1% significance level.

Pieces of evidence emerging from the stationarity test (Table 2) rectify that the enlisted series are integrated at levels *I(0)* and at order-one *I(1)*. This implies that the study relies on mixed-integrated series for empirical inferences. Unarguably, the above outcomes (mix integration), arising from both the augmented Dickey-Fuller (ADF) and the Lee-Strazicich procedures, eminently support the applications of the ARDL model. Meanwhile, the influence of structural breaks identified by the Lee-Strazicich nonlinear unit-root procedure will be largely accounted for by the application of the KRLS procedure, which is notable for revealing heterogeneous interference(s) of the explanatory variable(s) at various distributions of the explained variable. On the other hand, the cointegration test examined through the application of the Pesaran, Shin & Smith (2001) bounds test process (Table 3) signifies that capital migration in South Africa shares a long-run relationship with all the enclosed explanatory variables. Having established cointegration among the series, the study evaluates the specific long- and short-run impacts of each explanatory variable on the explained variable based on the ARDL and the quantile distributive impacts

based on the KRLS model. Accordingly, Tables 4 and 5 represent the empirical outcomes of the ARDL and KRLS estimations, respectively.

Table-4: ARDL estimates

Variable	Coefficient	t-Statistic	Probability
Panel A: Long-run effects			
Constant	44.46	0.73	0.47
$lnGPR_t$	-3.35	-3.47	0.00**
EPU_t	0.95	1.00	0.33
GRT_t	0.17	1.69	0.11
$lnOPN_t$	-2.05	-1.55	0.14
FDI_t	0.35	2.54	0.02**
$lnPOP_t$	-1.89	-0.26	0.79
Panel B: Short-run effects			
D(LNCF(-1))	0.51	4.70	0.00***
D(LNGPR)	-3.31	-6.07	0.00***
D(LNGPR(-1))	1.35	2.04	0.06*
D(EPU)	1.05	2.85	0.01**
D(EPU(-1))	1.37	2.78	0.01**
D(GRT)	0.15	2.91	0.01**
D(GRT(-1))	-0.10	-2.23	0.04**
D(FDI)	0.01	0.32	0.74
D(LNPOP(-1))	2.16	3.93	0.00***
ECT	-0.35	-0.50	0.00***
Panel C: Diagnostic tests			
Serial correlation	0.65(0.53)		
Heteroscedasticity	0.86(0.61)		
Ramsey-Reset	0.13(0.89)		
Normality	0.49(0.78)		
Adj-R^2	0.86		

Note: ***, ** and * indicate significant relationships at the 1%, 5% and 10% significance levels, respectively. Probability values of the diagnostic parameters are in parentheses.

On account of the estimated ARDL model (Table 4), the study rectifies that within the immediate time (short run), capital flight has an autoregressive function. This implies that the past volume of capital flight provides significant incentives for its current volume, hence making it a systemic problem. However, in the long run, the autoregressive effect becomes insignificant. This requires every effort to ensure a comprehensive eradication of capital migrations at all times. The investigation also reveals that GPR has a short-run significant negative influence on capital flight, whereas EPU has a significant positive short-run influence on capital flight. However, after some time lags, both GPR and EPU produced significant positive impacts on capital migration from South Africa. In specific terms, at lag 1, a percentage change in GPR and EPU increased the amount of capital migration by 1.35% and 1.37%, respectively. In the long run, GPR exerts a significant negative influence, while EPU exerts a nonsignificant positive influence on capital migration from South Africa. Accordingly, a 1% change in GPR ensures approximately -3.35% reductions in capital migrations, while EPU provides no remarkable long-run impacts. A similar outcome, an insignificant influence of EPU on capital outflows, is recorded in Uche & Effiom (2021a) in the case of Nigeria. Nevertheless, within the short run, the control variables (*GRT* and *lnPOP*) escalate capital migrations significantly from South Africa by approximately 0.15% and 2.16%, respectively. However, due to the influence of time (lag 1), capital flight declines by approximately -0.10% in response to a 1% change

in national income (GRT). Meanwhile, in the long run, neither national income (*GRT*) nor population (*lnPOP*) could have a remarkable influence on capital flight from South Africa. The above outcomes are at variance with the findings of Pradhan & Hiremath (2017) for India but validate the submissions of Anetor (2019).

FDI has an insignificant positive influence on capital migrations in the short run; however, the short-run outcome cannot be sustained in the long term; hence, capital flight increases significantly by approximately 0.35% in response to a 1% change in FDI net inflows. This largely implies that FDI is a significant long-run predictor of capital migration from South Africa. The significant positive impact of FDI on capital flight in South Africa also reflects the official revolving door hypothesis (Gankou et al, 2016). In this regard, policymakers should keep an eye on the inflows and outflows of FDI in South Africa to ensure drastic reductions in the outflows of illicit funds. Meanwhile, similar evidence is illustrated in Ndikumana & Sarr (2019). Further evidence suggests that trade openness (*lnTOP*) has no influence on capital flight from South Africa in the short run, whereas it produces insignificant reducing effects on capital flight in the long run. Meanwhile, the established outcomes of the ARDL model are judged robust in several respects, including the rightly signed and significant error-correction term (*ECT*), indicating a speed of adjustment of approximately -35% to equilibrium per annum. The reliability of the ARDL results is reinforced by the outcomes of the post estimation diagnostic tests, including the test for autocorrelation, heteroscedasticity and specification (Panel C in Table 4), as well as the normality and stability tests (Fig. 3). Accordingly, all the enlisted parameters of the postestimation analysis proved the reliability of the above empirical analysis.

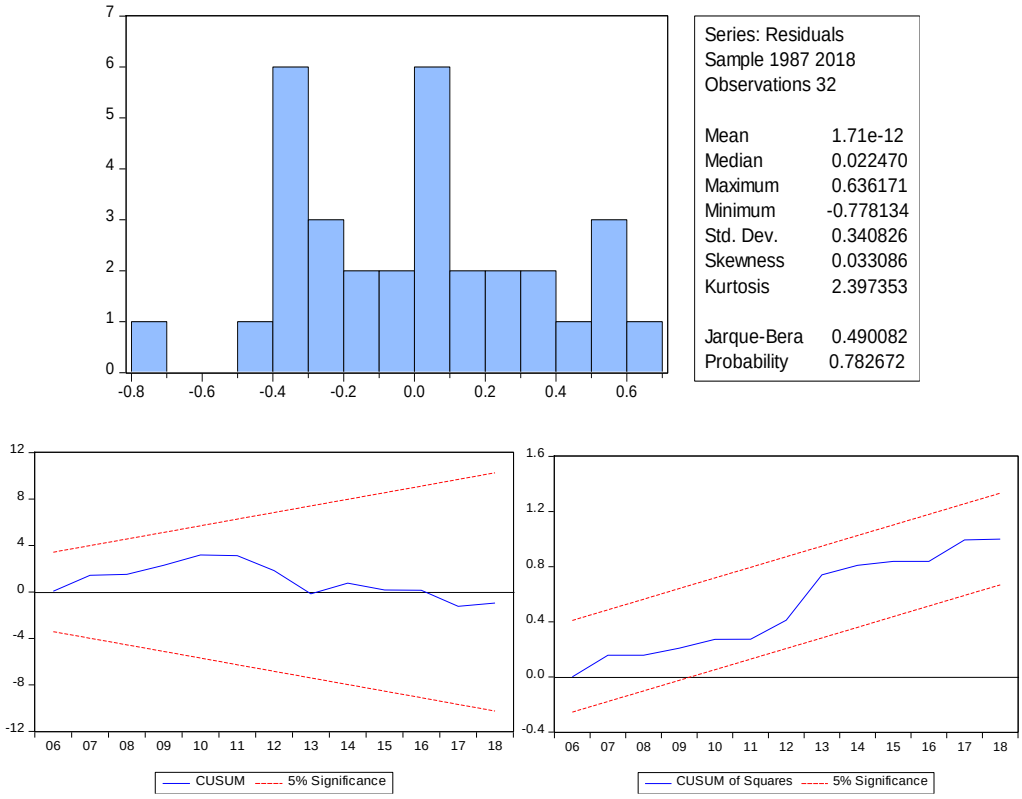


Fig. 3. Normality and stability graphs

Table-5: KRLS estimates

$lnCO_2$	Avg.	SE	t-Stat	P>|t|	P25	P50	P75
$lnGPR_t$	-1.19***	0.29	-3.99	0.00	-1.64	-1.28	-0.65
EPU_t	-0.08	0.24	-0.33	0.74	-0.45	-0.24	0.21
GRT_t	0.02	0.04	0.45	0.65	-0.04	0.02	0.09
$lnOPN_t$	0.67	0.48	1.41	0.17	-0.92	0.15	2.59
FDI_t	0.12*	0.06	1.85	0.07	0.07	0.12	0.28
$lnPOP_t$	-1.19	1.11	-1.07	0.29	-2.62	-0.97	0.30
Diagnostics test:							
Lambda	0.27	Tolerance	0.03	Sigma	6.00	Eff. Df	15.46
R^2	0.82	Looloss	25.13				

*Note: Avg. denotes average impacts P25, P50 and P75 represent the impacts of the explanatory variables on the regressand at the respective quantiles. *** and * indicate significant impacts at the 1% and 10% significance levels, respectively.*

As indicated earlier, the study adopted the KRLS nonlinear estimator for a confirmatory test and probed for the potential heterogeneous effects of GPR, EPU and all the listed control variables on capital migration from South Africa. The empirical outcome of the KRLS estimate is projected in Table 5. Accordingly, the KRLS result is, to a very large extent, consistent with the outcome of the ARDL test. This suggests that inferences arising from both estimates could at all times aid the enactment of policy strategies to mitigate capital migrations from South Africa. Accordingly, similar to the outcome of the ARDL model, but with little variation, GPR provides a significant negative influence on capital migration from South Africa at all times. Likewise, EPU has insignificant negative effects on capital migration in South Africa at all times. Whereas it produces insignificant positive effects based on the linear ARDL model, the nonlinear model (KRLS) indicates insignificant negative impacts. Overall, both the linear (ARDL) and nonlinear (KRLS) models indicate that EPU exhibits an insignificant impact on capital flight in South Africa. These outcomes suggest that the continuous migration of investible funds from South Africa is not a result of waves of geopolitical risk factors and economic policy uncertainties. In terms of the enlisted control variables, the KRLS estimate reveals that economic growth (*GRT*), trade openness (*lnTOP*), and population (*lnPOP*) are not significant predictors of capital migrations from South Africa both on average and across the quantile distributions of capital flight. However, such cannot be said of FDI net inflows, which provide positive and significant incentives for capital migrations from South Africa both on average and across all quantile distributions of capital flight. Specifically, a 1% change in FDI net inflows results in 0.12%, 0.07%, 0.12% and 0.28% increases in the amount of capital migrations on the average, 25th, 50th and 75th quantiles, respectively. This outcome, which largely agrees with the findings of Ndikumana & Sarr (2019), eminently contradicts the submissions of Ndikumana & Sarr (2016). Much as stated earlier, continuous scrutiny of FDI net inflows might provide the panacea for the mitigation of capital migrations from South Africa considering its all times positive and significant influence. Furthermore, the validity of the KRLS result is also reinforced by relevant diagnostic tests, including 82% overall fit (R^2), lambda (0.27), tolerance (0.03), Sigma (6.00) and Looloss (25.13).

4. Conclusions and policy briefs

The continuous migrations of the available investible funds from developing countries, in dire need of such funds, South Africa inclusive, to developed countries with very little need for such funds has been a burden on policymakers, governments and other relevant stakeholders, mostly those from countries where these funds are migrated. Several empirical enquires have been dedicated to unravelling the critical factors influencing this dastardly act. However, the panacea is still not in sight considering the divergent submissions of past enquires. Meanwhile, previous studies may have inadvertently failed to consider the influence of geopolitical risk factors (GPR) and economic policy uncertainties (EPU) on capital migrations

from South Africa, a notable economic giant on the African continent. Such relevant information may not have been ascertained for any other country based on available information. To remedy such a lacuna in knowledge, the current study explores the influence of GPR and EPU, as well as the moderating influence of economic growth, trade openness, FDI and population on illicit capital outflows from South Africa, considering that the empirical inferences arising therefrom could also benefit other developing countries in Africa and other continents.

From the estimated outcomes based on the two enhanced models (ARDL and KRLS), it is established that GPR provides significant reducing effects on capital migration from South Africa at all times. On its part, EPU provides no remarkable incentives for capital outflows from South Africa. The empirical outcomes also reveal that, in contrast to the revelations of most empirical studies, economic growth, trade openness and population were not significant predictors of capital flights in South Africa. However, the empirical outcomes arising from the two estimated models reveal the overriding positive and significant influence of FDI net inflows on capital migrations from South Africa. This suggests that FDI is a notable and remarkable conduit for capital migrations from South Africa.

In tune with the above outcomes, policymakers in South Africa should focus more on the inflows and outflows of FDI rather than geopolitical risk factors and economic policy uncertainties if they must win the war against capital migrations. It could also be helpful to ensure that such short-term significant positive influences of geopolitical risk factors and economic policy uncertainties, as well as economic growth and population on capital outflows, are curtailed in their entirety. As such, these funds could complement government developmental projects and further enhance the performance of the entire economy. Meanwhile, policymakers should not be unmindful of other potential negative impacts of GPR and EPU on economic and noneconomic fundamentals that were not considered in this investigation. Therefore, it is instructive that future studies incorporate some of these fundamentals to strengthen policy strategies. It is also instructive to extend the scope of this investigation up to the post-COVID-19 era.

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